Education Inequality Analysis

Team Neural Network, Data Science Spring 2026

The Difference Between Public, Private, and Charter Schools

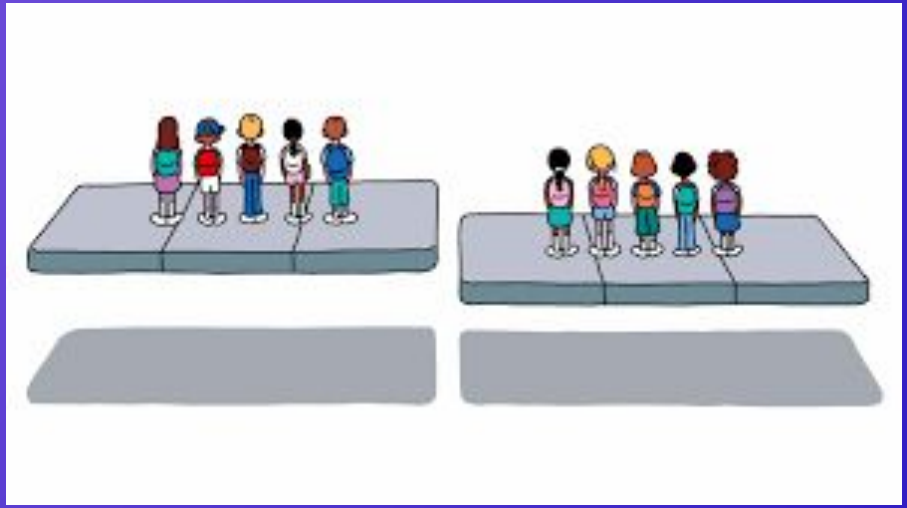
- Public: funded by taxpayers and public funds (local, state, and federal governments)
- Private: funded through endowments, tuition, and donations
- Charter: publicly funded, independently run (exempt from many typical school district requirements)

Annah

Objective

How do different factors like funding and school type affect educational outcomes?

- Analyze education inequality
- Examine funding, test scores, dropout rates
- Identify patterns



Education Inequality Data Set

Variables include:

- state
- type of school
- grade level
- funding per student
- average test score percent
- student teacher ratio
- low income percentage
- percentage of minority students
- internet access percent
- dropout rate percent

	id	school_name	state	school_type	grade_level	funding_per_student_usd	avg_test_score_percent	student_teacher_ratio	percent_low_income	percent_minority	internet_access_percent	dropout_rate_percent
0	1	Bowman High School	Michigan	Private	High	9575.80	65.9	17.5	30.0	44.0	74.1	3.44
1	2	Foster High School	Michigan	Charter	Middle	16733.10	87.3	24.2	93.5	40.1	79.9	8.67
2	3	Henson High School	New York	Public	Middle	24890.74	72.8	27.0	58.1	35.8	51.1	14.75
3	4	Weaver High School	Texas	Private	Elementary	6857.49	81.5	26.2	82.7	72.0	55.5	13.45
4	5	King Elementary School	Michigan	Public	Elementary	16910.73	44.7	20.2	41.4	16.1	57.8	6.98

Data Wrangling

- Loaded dataset into pandas df (Jupyter Notebook)
- Used Pandas, Matplotlib, Seaborn
- Checked structure (info(), describe())
- Verified no missing values or duplicates
- Filtered & grouped data (groupby())
- Sorted data for analysis (sort_values())

Data Structure

- ~1000 rows
- Multiple Feature Columns: funding, scores, school type, grade level
- Covers 10 states and 3 grade levels
- Structured for analysis and modeling
- Clean and well-formatted dataset

Data Types

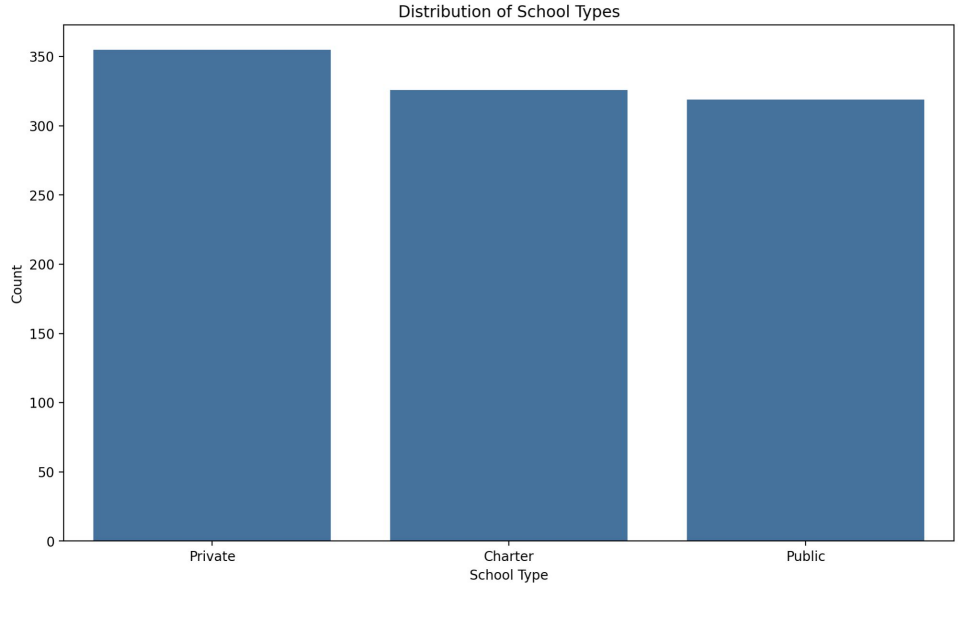
Categorical:

state, school type, grade level, etc

Numerical:

funding, scores, etc

Count Plot : School Type Distribution

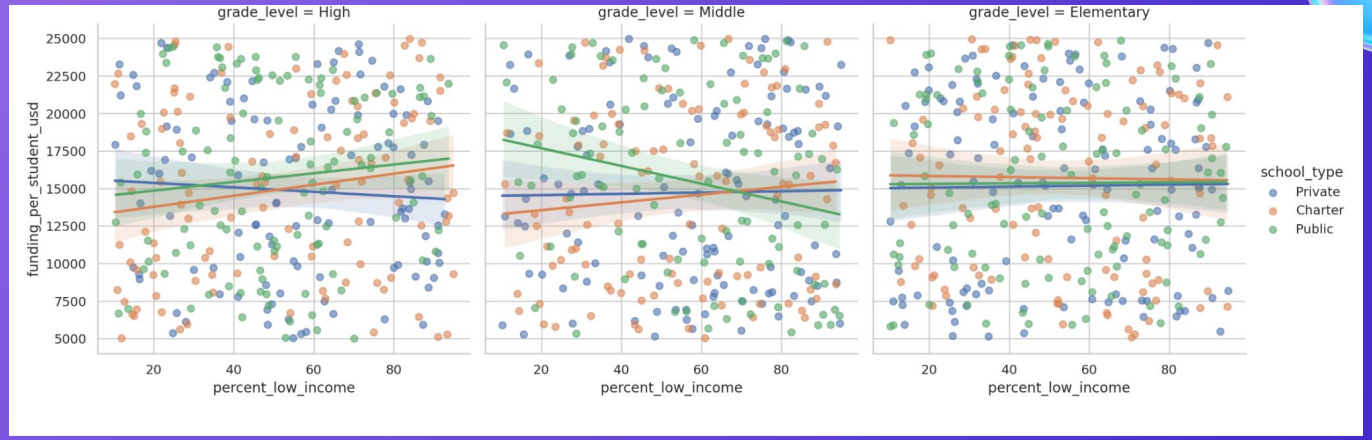


This count plot shows the distribution of schools by type, including public, private, and charter institutions.

- The visualization reveals a relatively balanced distribution among the three categories.
- Private schools appearing slightly more frequently.

This balance allows for meaningful comparisons across school types in further analysis, particularly when examining differences in funding, performance, and dropout rates.

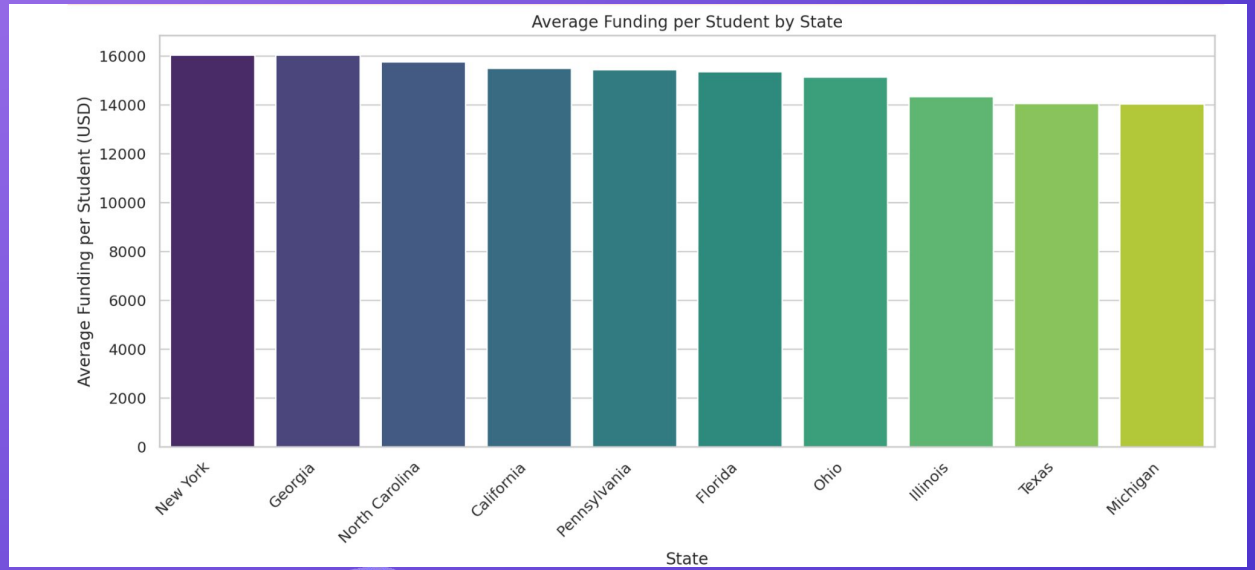
Funding against Grade Level



These scatterplots highlight the percentage of low income students and funding per student based on grade level across the data set.

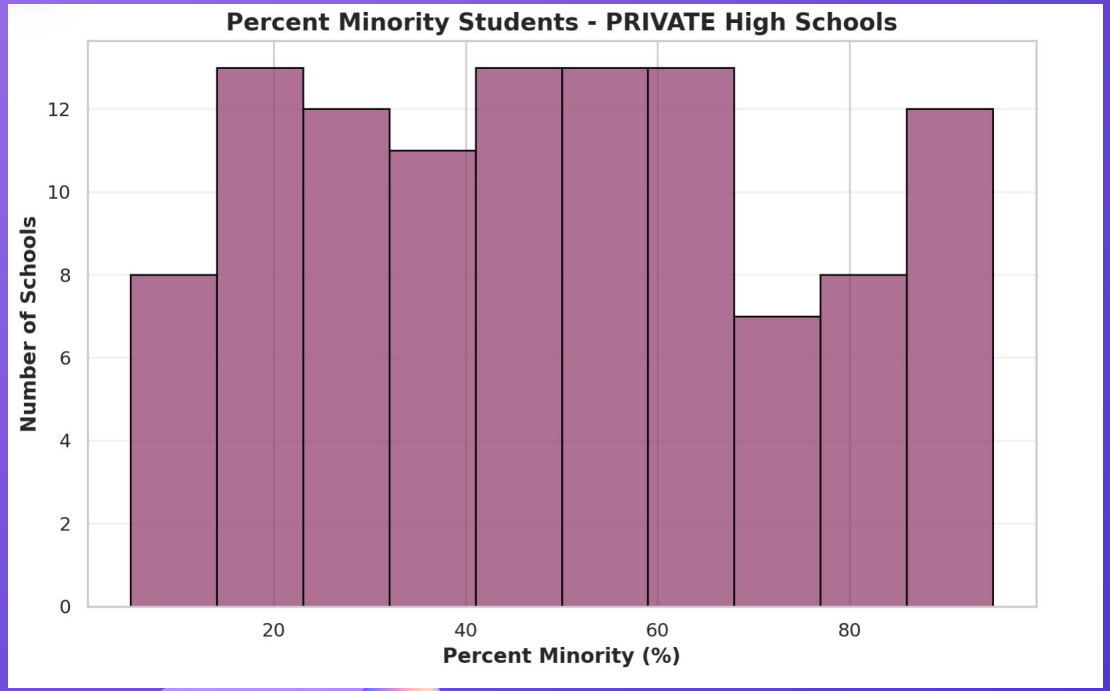
- For High School students, there is a strong positive correlation between these two variables for charter and public schools, and negative correlation for private schools.
- For Middle School, there is a strong negative correlation for public schools, positive correlation for charter schools, and no correlation for private schools.
- There is little to no correlation between the two variables for elementary schools.

Average Funding per Student by State



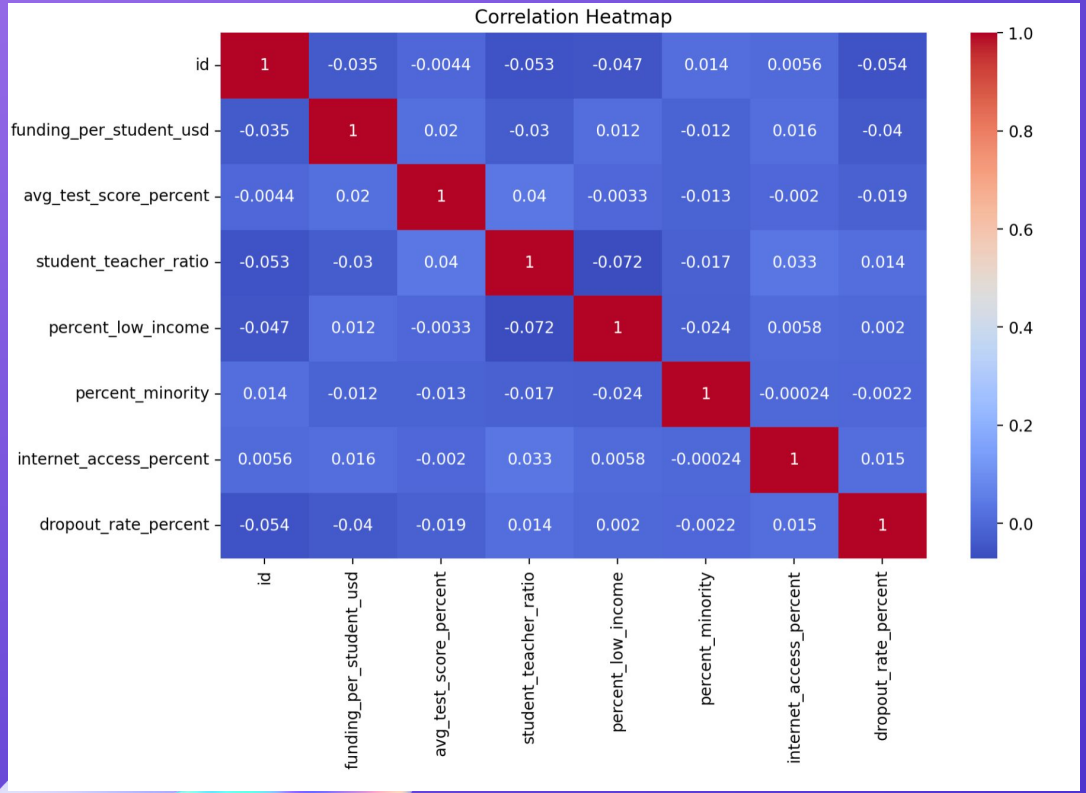
- This barplot highlights the average funding per student sorted by state. New York has the highest average funding per student, ranging at $160,000. Michigan and Texas have the lowest, ranging to about $140,000.

Percent Minority



- This histogram plot shows the percentage of minority students in private high schools across the 10 states in the data set.
- Most Private High schools have a minority population between 20 to 60%

Correlation Heat Map



- The heatmap shows that most variables have very weak correlations with each other, as the values are close to zero.
- This indicates that no single factor, such as funding or demographics, strongly predicts student outcomes on its own.
- Instead, educational outcomes are likely influenced by a combination of multiple factors.

Thank you!